WORD SEARCH PUZZLE SOLVER

A PROJECT REPORT

Submitted by

PRIYADHARSHINI S

8115U25AD081

in partial fulfillment of requirements for the award of the course

CGB1122 – DATA STRUCTURES

in

DEPARTMENT OF

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Under the guidance of

Mrs.M.KAVITHA

DEPARTMENT OF
ARTIFICIALINTELLIGENCEANDDATASCIENCE

K. RAMAKRISHNAN COLLEGE OF ENGINEERING (AUTONOMOUS)

SAMAYAPURAM, TRICHY – 621 112

Under

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BONAFIDE CERTIFICATE

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I hereby declare that the project report titled **"WORD SEARCH PUZZLE SOLVER"** is the result of original work done by me, to the best of my knowledge. This project report is submitted in partial fulfillment of the requirements for the award of the course **CGB1122 – DATA STRUCTURES** in the Department of Artificial interligiance and data science

Place: Samayapuram
Date: ……

Signature

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INSTITUTE VISION AND MISSION

VISION OF THE INSTITUTE:

To achieve a prominent position among the top technical institutions.

MISSION OF THE INSTITUTE:

M1: To bestow standard technical education par excellence through state of the art infrastructure, competent faculty and high ethical standards.

M2: To nurture research and entrepreneurial skills among students in cutting edge technologies.

M3: To provide education for developing high-quality professionals to transform the society.

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Engineering students will be able to:

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PO2. Problem Analysis: Identify and analyze complex engineering problems.

PO3. Design/Development of Solutions: Design solutions for engineering problems with safety and societal considerations.

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PO9. Individual and Team Work: Work effectively as an individual and in teams.

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PO11. Project Management and Finance: Apply management principles in projects.

PO12. Life-Long Learning: Engage in continuous learning and skill development.

ABSTRACT

A **Word Search Puzzle Solver** is a software-based solution designed to automatically identify and locate words hidden within a grid of letters. Word search puzzles are widely used for entertainment as well as educational purposes, where words may appear in various orientations such as horizontally, vertically, diagonally, and sometimes even in reverse directions. As the size and complexity of the puzzle increase, manually finding all the words becomes difficult and time-consuming.The proposed system aims to simplify this task by employing efficient search algorithms and systematic grid traversal techniques. The solver takes a letter grid and a list of target words as input, and processes the grid by scanning each cell in all possible directions to detect matching character sequences. Advanced techniques such as backtracking or optimized string-matching methods can be used to improve the speed and accuracy of the search process.This project emphasizes key concepts in computer science, including matrix representation, pattern recognition, and algorithm design. It can be implemented using programming languages such as C++ or Python, making it suitable for academic learning and practical applications. Additionally, the system can be enhanced with features like a graphical user interface, real-time solving, and support for larger and more complex puzzles.Overall, the Word Search Puzzle Solver provides an efficient, accurate, and user-friendly approach to solving word search puzzles, while also serving as a valuable educational tool for understanding computational problem-solving techniques.

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LIST OF ABBREVIATIONS

ABBREVIATIONS	FULL FORM
CLI	COMMAND LINE INTERFACE
GUI	GRAPHICAL USER INTERFACE
API	APPLICATION PROGRAMMING INTERFACE
RAM	RANDOM ACCESS MEMORY
CPU	CENTRAL PROCESSING UNIT
IP	INTERNET PROTOCOL
TCP	TRANSMISSION CONTROL PROTOCOL
UDP	USER DATAGRAM PROTOCOL
ICMP	INTERNET CONTROL MESSAGE PROTOCOL
HTTP	HYPERTEXT TRANSFER PROTOCOL
QoS	QUALITY OF SERVICE
IDS	INTRUSION DETECTION SYSTEM

CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

A **Word Search Puzzle Solver** is a tool or program designed to automatically find hidden words within a grid of letters. Word search puzzles are a popular form of entertainment and educational activity, where words are placed horizontally, vertically, or diagonally in a grid. Solving these puzzles manually can sometimes be time-consuming, especially for larger grids with complex word placements.

The purpose of a word search puzzle solver is to simplify this process by using efficient algorithms to quickly locate given words in the puzzle. It scans the grid in all possible directions and matches sequences of letters with the target words. Such solvers are useful for students, puzzle enthusiasts, and developers learning problem-solving techniques in computer science.

Overall, a word search puzzle solver demonstrates concepts like string matching, matrix traversal, and algorithm optimization, making it both a practical and educational application.

1.2 PURPOSE AND IMPORTANCE

A word search puzzle solver is a tool designed to automatically identify hidden words within a grid of letters, where words may appear in various directions such as horizontal, vertical, diagonal, forward, or backward. The main purpose of this solver is to make the process of finding words faster and easier, especially in large or complex puzzles. It helps users save time and effort by quickly scanning the grid and ensuring that no words are missed. This is particularly useful for beginners, students, or anyone who finds manual searching challenging.

The importance of a word search puzzle solver extends to both educational and practical uses. It helps improve vocabulary, spelling, and pattern recognition skills, making it valuable for learning purposes. Additionally, it enhances logical thinking and problem-solving abilities by demonstrating systematic search methods. The solver is also beneficial for puzzle creators and game developers, as it can be used to test and verify puzzles efficiently. Overall, it makes puzzle-solving more accessible, accurate, and enjoyable for a wide range of users.

1.3 OBJECTIVES

- To **automatically find hidden words** in a letter grid in all possible directions.
- To **reduce time and manual effort** required to solve complex puzzles.
- To **improve accuracy** by ensuring no words are missed.
- To **assist users in learning vocabulary and spelling** through interactive solving.
- To **enhance logical thinking and pattern recognition skills**.
- To **support beginners and students** in understanding puzzle-solving techniques.
- To **help developers test and validate puzzle designs** efficiently.

1.4 PROJECT SUMMARIZATION

The Word Search Puzzle Solver is a software-based project developed to automatically find hidden words within a grid of letters. The system is designed to search words in all possible directions, including horizontal, vertical, diagonal, forward, and backward. By using efficient algorithms, the solver quickly scans the puzzle and identifies the required words with high accuracy. This reduces the time and effort needed for manual solving, especially in large or complex puzzles.

This project is useful in both educational and recreational fields. It helps students improve vocabulary, spelling, and pattern recognition skills while also enhancing

logical thinking and problem-solving abilities. Additionally, it can be used by puzzle creators and game developers to design and verify puzzles effectively. Overall, the Word Search Puzzle Solver provides a fast, accurate, and user-friendly solution, making puzzle-solving more efficient and enjoyable.

Key Benefits of This Project:

- **Time-saving**: Quickly finds words, reducing the effort compared to manual searching.
- **High accuracy**: Ensures all hidden words are identified without missing any.
- **Improves learning**: Helps enhance vocabulary, spelling, and pattern recognition skills.
- **Enhances problem-solving**: Encourages logical thinking and systematic search methods.
- **User-friendly**: Useful for beginners as well as experienced users.
- **Supports development**: Assists puzzle creators in designing and testing puzzles efficiently.
- **Versatile use**: Can be applied in education, games, and recreational activities

CHAPTER 2
PROJECT METHODOLOGY

2.1 INTRODUCTION TO SYSTEM ARCHITECTURE

System architecture refers to the overall structure and design of a system, showing how different components interact and work together to achieve a specific goal. It provides a high-level view of the system, including hardware, software, data flow, and communication between modules. The main purpose of system architecture is to ensure that all parts of the system are well-organized, efficient, and capable of meeting the required performance and functionality.

2.1.1 High-Level System Architecture

The high-level system architecture of a Word Search Puzzle Solver describes the main components of the system and how they interact to perform the task efficiently. It is typically divided into three major layers: input, processing, and output.

2.1.2 Components of the System Architecture

a. Input component: The **input component** is responsible for accepting the puzzle grid and the list of words from the user, either through manual entry or file upload.

b. Processing component: The processing component is the core part of the Word Search Puzzle Solver, where the actual word searching operation takes place. Its main function is to scan the given grid of letters and locate the specified words in all possible directions, such as horizontal, vertical, and diagonal, both forward and backward. This component uses efficient searching algorithms to systematically check each position in the grid and compare it with the target words, ensuring that all matches are found accurately.

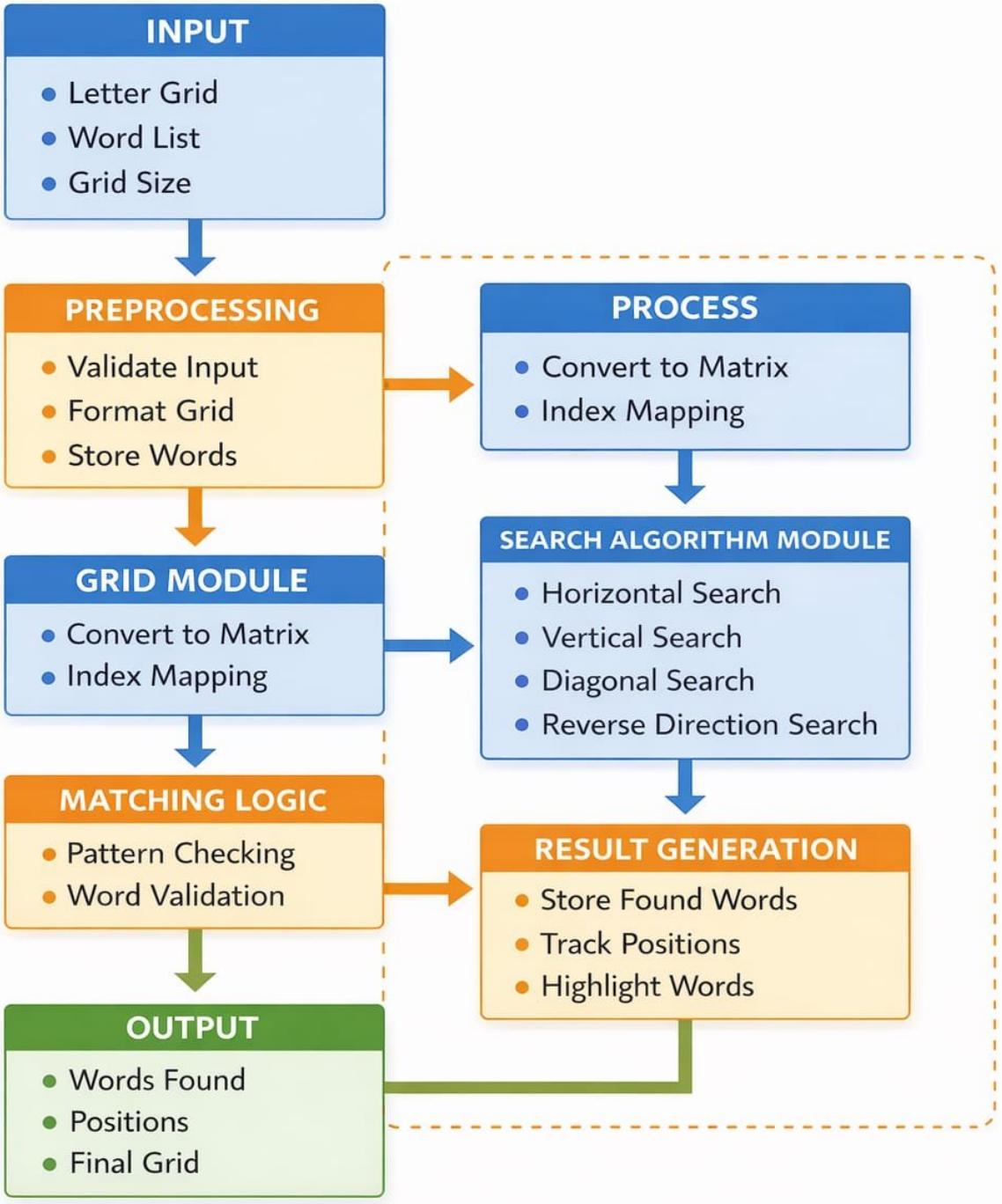
c. Output component: The output component is responsible for presenting the results of the Word Search Puzzle Solver to the user in a clear and understandable format. After the processing component identifies the hidden words, the output module displays the results by listing the found words along with their positions in the grid. It may also highlight the words directly within the puzzle grid, making it easier for users to visualize where each word is located.

2.2 DETAILED SYSTEM ARCHITECTURE DIAGRAM

The system starts with the **user input**, where the puzzle grid and word list are provided. This data moves to the **preprocessing unit**, which cleans and validates the input to ensure consistency. Then, the **processing unit** applies algorithms to search for words in all directions. Once words are found, the **result generator** organizes the results with positions and directions. Finally, the **output interface** displays the results clearly, sometimes highlighting the words in the grid or allowing users to save the output.

This architecture ensures the system is **efficient, organized, and easy to understand**, while also supporting future improvements.

Architecture Diagram: (INPUT- OUTPUT)



CHAPTER 3
DATA STRUCTURE PREFERENCE

3.1 EXPLANATION OF WHY 2D ARRAYS WERE CHOSEN

A **2D array** is the most suitable data structure for representing a word search puzzle because the puzzle itself is organized in a grid format consisting of rows and columns. Each cell in the grid contains a single character, and a 2D array provides a direct and natural way to store and access these characters using row and column indices.

The use of a 2D array allows **constant-time access (O(1))** to any element in the grid, which is essential for efficient searching. Since the word search problem requires checking adjacent cells in multiple directions—such as horizontal, vertical, and diagonal—the 2D array structure makes it easy to navigate by simply modifying the row and column indices.

Additionally, the 2D array works effectively with search algorithms like **Depth First Search (DFS)** or **Backtracking**, where traversal to neighboring cells is required. Boundary conditions can also be handled easily by checking the limits of row and column indices.

Compared to other data structures, a 2D array is **simple, memory-efficient, and easy to implement**, making it the preferred choice for grid-based problems like word search puzzles.

3.2 COMPARISON WITH OTHER DATA STRUCTURES

Data structure	Advantages	Disadvantages	Suitability
2D Array	Simple representation of grid, O(1) access, easy traversal in all directions	Fixed size, less flexible	✓ Highly suitable (best for grid)
1D Array	Simple, O(1) access	Hard to represent grid, complex index calculation	✗ Not suitable
Linked List	Dynamic size, flexible memory	Slow access O(n), difficult navigation	✗ Not suitable
Graph	Flexible structure, models relationships	Complex implementation, extra memory needed	⚠ Less suitable (overhead)
Hash Table	Fast lookup O(1)	No positional information, unordered	✗ Not suitable
Trie (Tree)	Efficient word storage, fast prefix search	Extra memory usage, cannot represent grid	⚠ Suitable only for word storage

3.3 ADVANTAGES AND DISADVANTAGES OF USING 2D ARRAY

3.3.1 Advantages:

- Natural representation of grid (rows and columns)
- Direct access to elements using indices
- Fast access time **O(1)**
- Easy traversal in all directions
- Simple to implement and understand
- Efficient memory usage
- Works well with DFS and backtracking
- Easy boundary checking
- Reduces complexity in grid-based problems

3.3.2 Disadvantages:

- Fixed size (cannot be resized dynamically)
- May lead to memory wastage if grid is not fully used
- Not flexible for irregular or non-rectangular data
- Insertion and deletion operations are difficult
- Requires contiguous memory allocation
- Less efficient for very large datasets
- Cannot store complex relationships like graphs
- Limited to structured grid-based problems

CHAPTER 4
DATA STRUCTURES METHODOLOGY

4.1 2D ARRAY USAGE

In a word search puzzle solver, a 2D array is used to represent the puzzle grid in the form of rows and columns, where each cell contains a single character. This structure allows the solver to easily access any letter using its row and column position. The 2D array makes it possible to systematically traverse the grid and consider each cell as a starting point for searching words.

It also supports movement in multiple directions such as horizontal, vertical, and diagonal by simply adjusting the row and column indices. This is important because words in a puzzle can appear in any direction. Additionally, the 2D array enables efficient implementation of search algorithms like Depth First Search (DFS), where neighboring cells are explored step by step.

Overall, the 2D array provides a simple, efficient, and structured way to store and process the puzzle grid, making it the most suitable choice for implementing a word search puzzle solver.

4.1.1 Depth First Search(DFS)

Depth First Search (DFS) is a searching technique used in a word search puzzle solver to explore all possible paths in the grid to find a given word. It starts from a particular cell and explores as far as possible along one direction before backtracking and trying other directions.

In a word search puzzle, DFS begins at each cell of the **2D array** and checks whether the first letter of the word matches the cell value. If it matches, the algorithm continues searching the next letters by moving to adjacent cells such as

up, down, left, right, and sometimes diagonally. During this process, each visited cell is temporarily marked to avoid revisiting the same cell in the current path.

If the sequence of characters matches the given word, the search is successful. If at any point the characters do not match, the algorithm **backtracks** to the previous cell and explores other possible directions. This process continues until the word is found or all possibilities are exhausted.

DFS is effective because it systematically explores all possible paths in the grid, ensuring that no valid word formation is missed.

4.1.2 Trie (Prefix Tree)

A **Trie (Prefix Tree)** is a tree-based data structure used in a word search puzzle solver to store a list of words efficiently. It is especially useful when searching for multiple words in a grid because it allows **fast prefix checking** and reduces unnecessary comparisons.

In a word search puzzle solver, all the given words are first inserted into the Trie. Each node in the Trie represents a character, and a path from the root to a node represents a prefix of a word. While performing the search (usually using DFS on the 2D array), the algorithm checks whether the current sequence of characters matches any prefix in the Trie.

If the current path does not match any prefix in the Trie, the search is stopped immediately (pruning), which saves time. If a complete word is found (end of a Trie branch), it is added to the result. This process continues until all possible paths in the grid are explored.

Thus, the Trie helps in optimizing the search by avoiding unnecessary explorations and improving overall efficiency.

4.2 NODE STRUCTURES

In a word search puzzle solver, the **node structure** is mainly used in the **Trie (Prefix Tree)** to store and search words efficiently.

4.2.1 Trie Node Structures

Each node represents a single character and contains the following components:

1.children links

- Stores references to the next characters
- Typically an array of size **26** (for a–z)

2. End of Word flag

- A boolean value (`isEndOfWord`)
- Indicates whether a complete word ends at this node

3. Word storage(optional)

- Stores the full word when it is completed
- Helps in directly adding the word to the result

- **How its work in solver**

- Words are inserted into the Trie using this node structure
- During DFS on the 2D grid:
    - Move from one node to another based on characters
    - If no child exists → stop search (pruning)
    - If `isEndOfWord = true` → word found

4.3 INITIALIZATION, INSERTION & DELETION

4.3.1 Initialization

Initialization is the first step in designing a word search puzzle solver, where all required data structures are created and prepared before the searching process begins.

In this stage, the puzzle grid is initialized using a **2D array**, where each element stores a character of the grid. This allows easy access to any cell using row and column indices. Next, the list of words to be searched is initialized and stored in a suitable data structure such as a **Trie**, which supports efficient prefix-based searching.

4.3.2 Insertion

Insertion is the process of adding words into the **Trie data structure** before starting the search in the puzzle. This step is important because it organizes all the given words in a way that allows fast and efficient searching.

During insertion, each word is processed **character by character**. Starting from the root node, the algorithm checks whether a node for the current character already exists. If it does not exist, a new node is created. The process then moves to that node and continues with the next character of the word. This continues until all characters of the word are inserted.

After inserting the last character, a special flag called **end-of-word** is marked as true. This indicates that a complete word ends at that node. By doing this for all words, the Trie stores them in a prefix-based structure, where common prefixes are shared among multiple words.

This insertion method reduces memory usage and improves search efficiency, because during the solving process, invalid paths can be stopped early if they do not match any prefix in the Trie

4.3.3 Deletion

Deletion in a Trie refers to the process of removing a word from the data structure without affecting other words stored in it. This operation must be handled carefully because multiple words may share common prefixes.

During deletion, the algorithm first traverses the Trie following the characters of the word from the root node. Once it reaches the last character of the word, it **unmarks the end-of-word flag**, indicating that the word is no longer present in the Trie.

After unmarking, the algorithm checks whether the nodes used by the word are shared with other words. If a node has no children and is not marked as the end of another word, it can be safely deleted. This process continues backward (backtracking) to remove unnecessary nodes while preserving nodes that are part of other words.

In a word search puzzle solver, deletion is sometimes used to **avoid duplicate results** by removing a word once it is found, thereby improving efficiency.

CHAPTER 5
MODULES

5.1 INPUT MODULE

The input module is responsible for collecting the required data for the solver. It takes the puzzle grid and the list of words that need to be searched. The input can be provided by the user, read from a file, or generated by the system. This module ensures that the data is properly formatted and ready for further processing. Proper validation may also be performed to avoid errors during execution.

5.2 GRID INITIALIZATION MODULE

This module initializes the puzzle grid using a **2D array**. Each element of the array stores a single character from the puzzle. The grid structure is created in terms of rows and columns, allowing easy access to any position using indices. This module plays a crucial role in representing the puzzle in a structured format suitable for traversal and searching.

5.3 TRIE CONSTRUCTION MODULE

In this module, all the words to be searched are inserted into a **Trie (Prefix Tree)**. The Trie organizes words in a hierarchical structure based on their prefixes. Each character of a word is stored as a node, and the end of each word is marked with a special flag. This module helps in improving search efficiency by allowing quick prefix matching and reducing unnecessary comparisons.

5.4 SEARCH MODULE

The search module is the core component of the solver. It uses **Depth First Search (DFS)** to explore the grid. Starting from each cell of the 2D array, the algorithm checks whether the character matches the beginning of any word stored in the Trie.

If it matches, the search continues by exploring adjacent cells in different directions such as up, down, left, right, and diagonals. This module ensures that all possible paths are explored systematically.

5.5 VISTED TRACKING MODULE

This module is used to keep track of cells that have already been visited during the current search path. It typically uses a **boolean 2D array** or a set. This prevents the algorithm from revisiting the same cell within a single word search, which could lead to incorrect results or infinite loops. Once the search moves back (backtracking), the visited status is reset for further exploration

5.6 BACKTRACKING MODULE

Backtracking is an essential part of the search process. This module ensures that after exploring one path, the solver returns to the previous state and tries other possible paths. It helps in exploring all combinations without permanently altering the grid or visited states. Backtracking improves correctness and ensures that no valid word is missed.

5.7 OUTPUT MODULE

The output module is responsible for presenting the final results to the user. It displays the list of found words and may also include additional information such as their positions or paths in the grid. The output can be shown on the screen, written to a file, or passed to another system for further processing.

CHAPTER 6
CONCLUSION & FUTURE SCOPE

6.1 CONCLUSION

The word search puzzle solver is an efficient system designed to locate words within a grid of characters using appropriate data structures and algorithms. By representing the puzzle as a **2D array**, the solver can easily access and traverse the grid in all possible directions. The use of advanced structures like the **Trie (Prefix Tree)** further enhances performance by enabling fast and efficient prefix-based searching.

The integration of **Depth First Search (DFS)** with backtracking ensures that all possible paths are explored systematically while avoiding unnecessary computations. Additional components such as visited tracking and result storage help maintain accuracy and organize the output effectively.

Overall, the word search puzzle solver provides a structured and optimized approach to solving grid-based problems. It demonstrates how combining suitable data structures with efficient algorithms can significantly improve performance and scalability, making it a practical solution for both small and large puzzles.

6.2 FUTURE SCOPE

The word search puzzle solver has a wide scope for future improvements and applications as technology advances. One major enhancement is the integration of **Artificial Intelligence (AI) and Machine Learning (ML)** techniques to improve pattern recognition and make the solver more intelligent in predicting word placements and optimizing search paths.

In the future, the system can be extended to support **real-time interactive applications**, such as mobile apps and educational games, where users can play and solve puzzles dynamically. It can also be improved to handle **larger and more complex grids** efficiently by using advanced optimization techniques and parallel processing.

Another important scope is the development of **multilingual support**, allowing the solver to work with different languages and character sets. Additionally, integration with **voice input and gesture-based interfaces** can make the system more user-friendly and accessible.

The solver can also be applied in **educational tools**, competitive programming platforms, and cognitive skill development applications. With further enhancements, it can be used in **game development, data analysis, and AI-based puzzle generation systems**.

Overall, the future scope of the word search puzzle solver is vast, with opportunities for making it more intelligent, interactive, and scalable.

APPENDIX A – SOURCE CODE

Program: Word search puzzle solver(C Program)

```
#include <stdio.h>
#include <string.h>

#define ROWS 8
#define COLS 8
#define MAX_WORDS 15

// Predefined grid
char grid[ROWS][COLS] = {
    {'C','A','T','F','G','H','I','J'},
    {'B','G','E','S','R','Q','W','X'},
    {'I','T','A','E','L','M','N','O'},
    {'S','O','M','E','N','P','A','B'},
    {'D','O','G','X','Y','Z','C','D'},
    {'H','O','U','S','E','K','L','M'},
    {'P','E','N','C','I','L','Q','R'},
    {'S','U','N','T','V','W','X','Y'}
};

// Directions
int dx[] = {0, 0, 1, -1, 1, 1, -1, -1};
int dy[] = {1, -1, 0, 0, 1, -1, 1, -1};

char *dirName[] = {
    "RIGHT", "LEFT", "DOWN", "UP",
    "DOWN-RIGHT", "DOWN-LEFT", "UP-RIGHT", "UP-LEFT"
};

// Word list inside program
```

```
char words[MAX_WORDS][20] = {
    "CAT","DOG","SOME","TEA","HOUSE",
    "PENCIL","SUN","EAT","GO","HOME",
    "FISH","BOOK","NOTE","CODE","JAVA"
};

int totalWords = 15;

// Print grid
void printGrid() {
    printf("\n==== PUZZLE GRID ====\n");
    for(int i=0;i<ROWS;i++) {
        for(int j=0;j<COLS;j++) {
            printf("%c ", grid[i][j]);
        }
        printf("\n");
    }
}

// Search from a position
int searchFrom(char word[], int row, int col, int *direction) {
    int len = strlen(word);

    if(grid[row][col] != word[0])
        return 0;

    for(int dir = 0; dir < 8; dir++) {
        int k, newRow = row, newCol = col;

        for(k = 1; k < len; k++) {
            newRow += dx[dir];
            newCol += dy[dir];
```

```
            if(newRow < 0 || newRow >= ROWS || newCol < 0 || newCol >= COLS)
                break;

            if(grid[newRow][newCol] != word[k])
                break;
        }

        if(k == len) {
            *direction = dir;
            return 1;
        }
    }
    return 0;
}

// Find word
void findWord(char word[]) {
    int found = 0;

    printf("\n----------------------------------\n");
    printf("Searching: %s\n", word);
    printf("----------------------------------\n");

    for(int i=0;i<ROWS;i++) {
        for(int j=0;j<COLS;j++) {
            int dir;

            if(searchFrom(word, i, j, &dir)) {
                printf("FOUND at (%d,%d) Direction: %s\n",
                    i, j, dirName[dir]);
                found = 1;
```

```
            }
        }
    }

    if(!found)
        printf("NOT FOUND\n");
}

int main() {

    printf("========================================\n");
    printf(" WORD SEARCH PUZZLE SOLVER (STATIC GRID)\n");
    printf("========================================\n");

    printGrid();

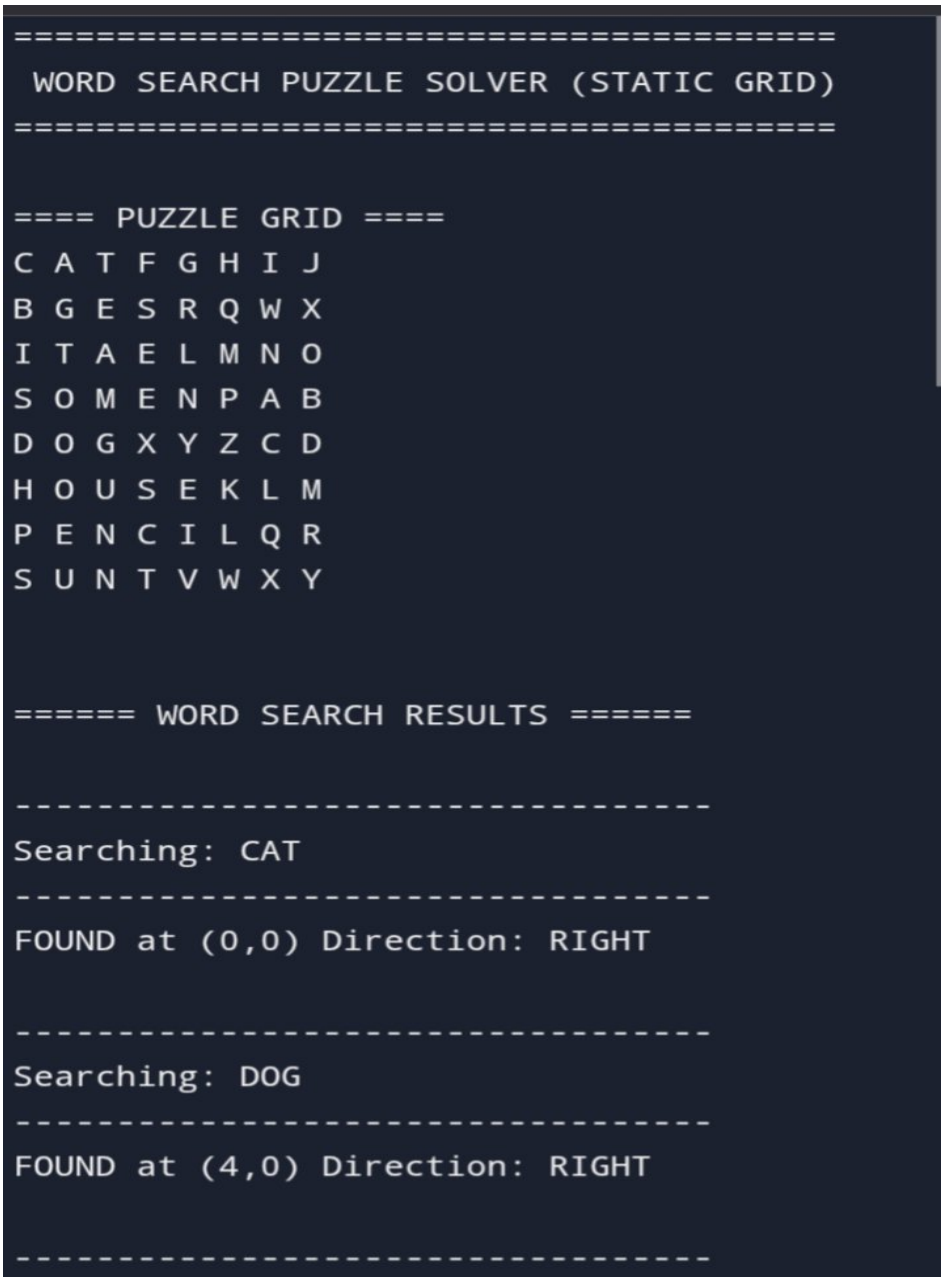
    printf("\n\n====== WORD SEARCH RESULTS ======\n");

    for(int i=0;i<totalWords;i++) {
        findWord(words[i]);
    }

    printf("\n========= END =========\n");

    return 0;
}
```

APPENDIX B – SCREENSHOTS



```
Searching: SOME
----------------------------------
FOUND at (3,0) Direction: RIGHT

----------------------------------
Searching: TEA
----------------------------------
FOUND at (0,2) Direction: DOWN

----------------------------------
Searching: HOUSE
----------------------------------
FOUND at (5,0) Direction: RIGHT

----------------------------------
Searching: PENCIL
----------------------------------
FOUND at (6,0) Direction: RIGHT

----------------------------------
Searching: SUN
----------------------------------
FOUND at (7,0) Direction: RIGHT

----------------------------------
Searching: EAT
----------------------------------
FOUND at (2,3) Direction: LEFT
```

```
Searching: GO
----------------------------------
FOUND at (4,2) Direction: LEFT

----------------------------------
Searching: HOME
----------------------------------
FOUND at (5,0) Direction: UP-RIGHT

----------------------------------
Searching: FISH
----------------------------------
NOT FOUND

----------------------------------
Searching: BOOK
----------------------------------
NOT FOUND

----------------------------------
Searching: NOTE
----------------------------------
NOT FOUND

----------------------------------
Searching: CODE
----------------------------------
NOT FOUND
```

```
----------------------------------
Searching: JAVA
----------------------------------
NOT FOUND

========= END =========
```

REFERENCES

1. **Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran**
   "Fundamentals of Data Structures in C"
   → Used for understanding arrays, recursion, trees, graphs, and DFS techniques.
2. **Reema Thareja**
   "Data Structures Using C"
   → Helps in learning implementation of arrays, stacks, queues, and tree structures like Trie.
3. **Mark Allen Weiss**
   "Data Structures and Algorithm Analysis in C"
   → Provides algorithm design techniques and complexity analysis for efficient searching.
4. **Brian W. Kernighan & Dennis M. Ritchie**
   "The C Programming Language"
   → Reference for C programming concepts used in implementation.
5. **GeeksforGeeks**
   https://www.geeksforgeeks.org
   → Used for understanding Trie, DFS, backtracking, and word search problem solutions.